FinRAG — results against HDRR (2026)

Haroon M Bilal · Muhammad Bin Sarwar · 28 October 2026

> On 258 held-out FinDER queries scored by Cheng et al.'s own GPT-4.1 judge, using their rubric and prompt verbatim, FinRAG reaches 6.02/10 against HDRR's 7.64.

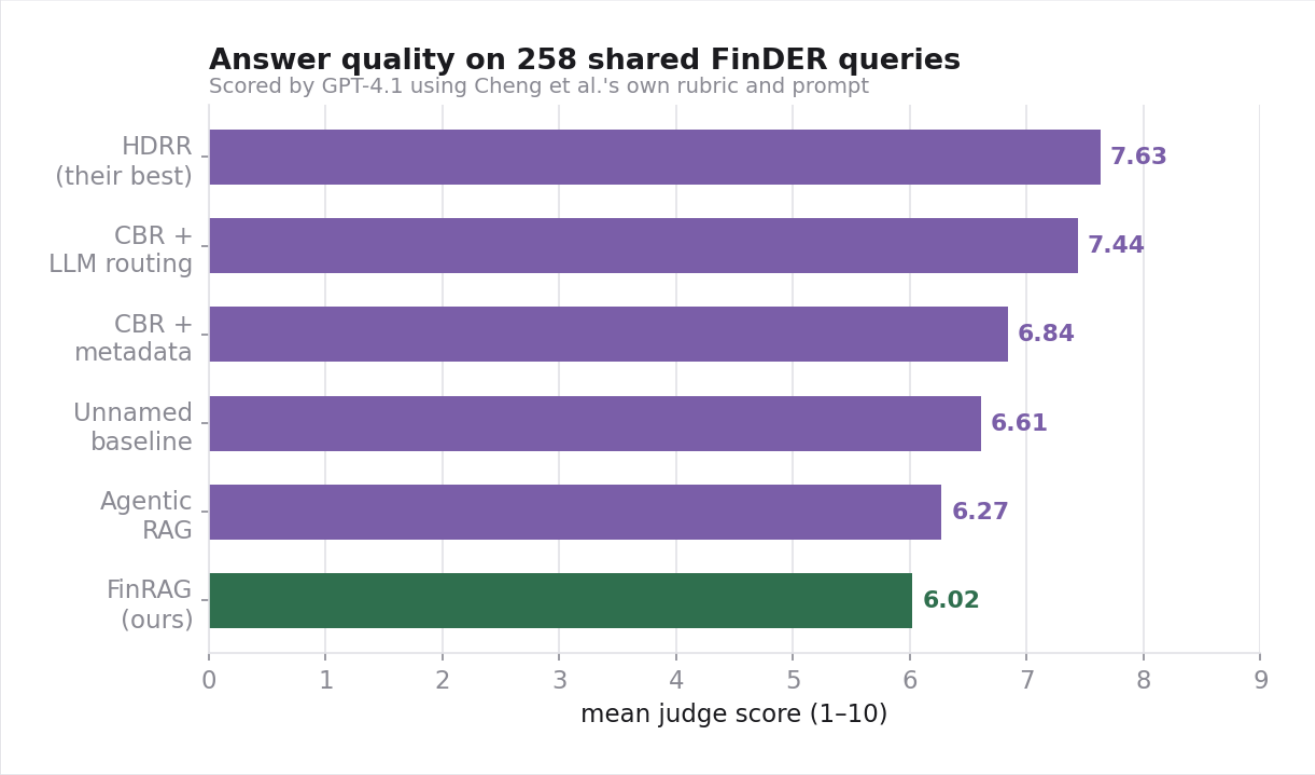
> The grader was verified against 40 of their published answers: 90% exact agreement, 100% within one point, r = 0.993. The comparison is therefore like-for-like.

> The 1.61-point gap decomposes into 0.67 from retrieval and 0.93 from generator capacity. On retrieval itself — which HDRR never reports — FinRAG reaches NDCG@10 0.599, and a zero-cost router matches their per-query LLM router's decision on 100% of queries where both commit.

All queries come from filings never seen during training. Confidence intervals are 10,000-sample paired bootstraps over identical queries.

FIGURE 1

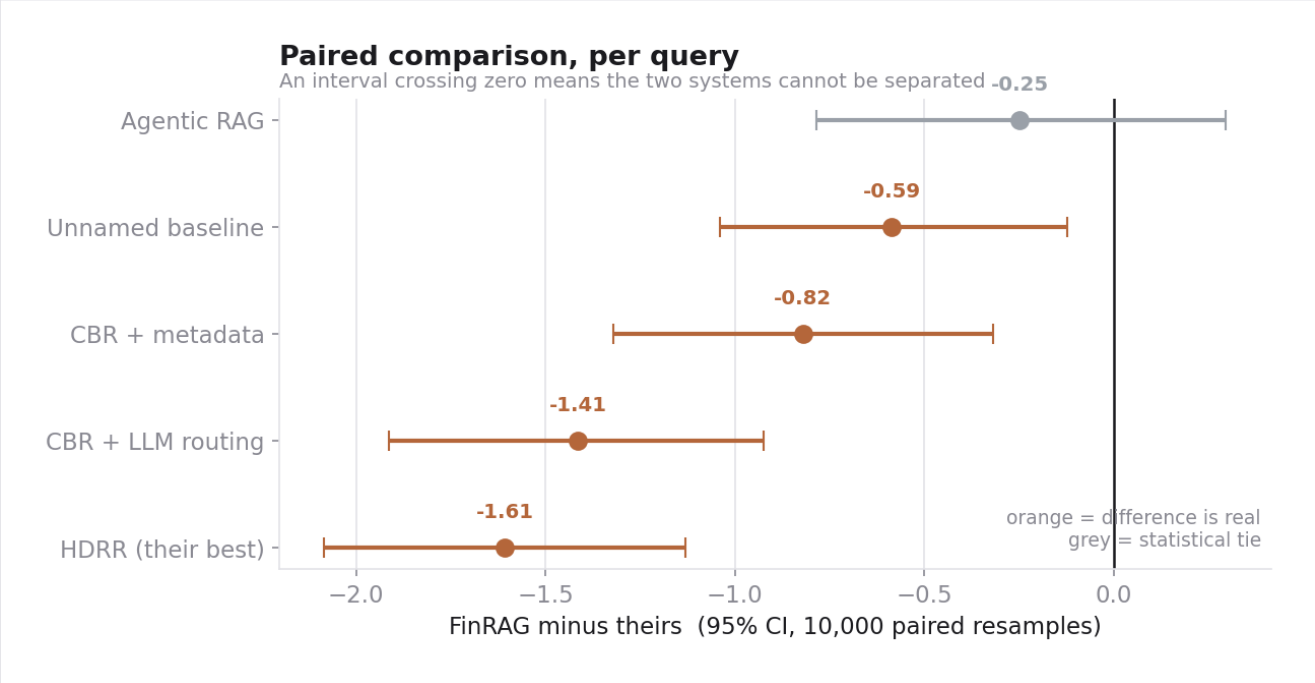
Answer quality against every published system



Same 258 held-out queries, same judge, same rubric. FinRAG runs entirely on one laptop GPU; every other system generates its answers with a frontier API model.

FIGURE 2

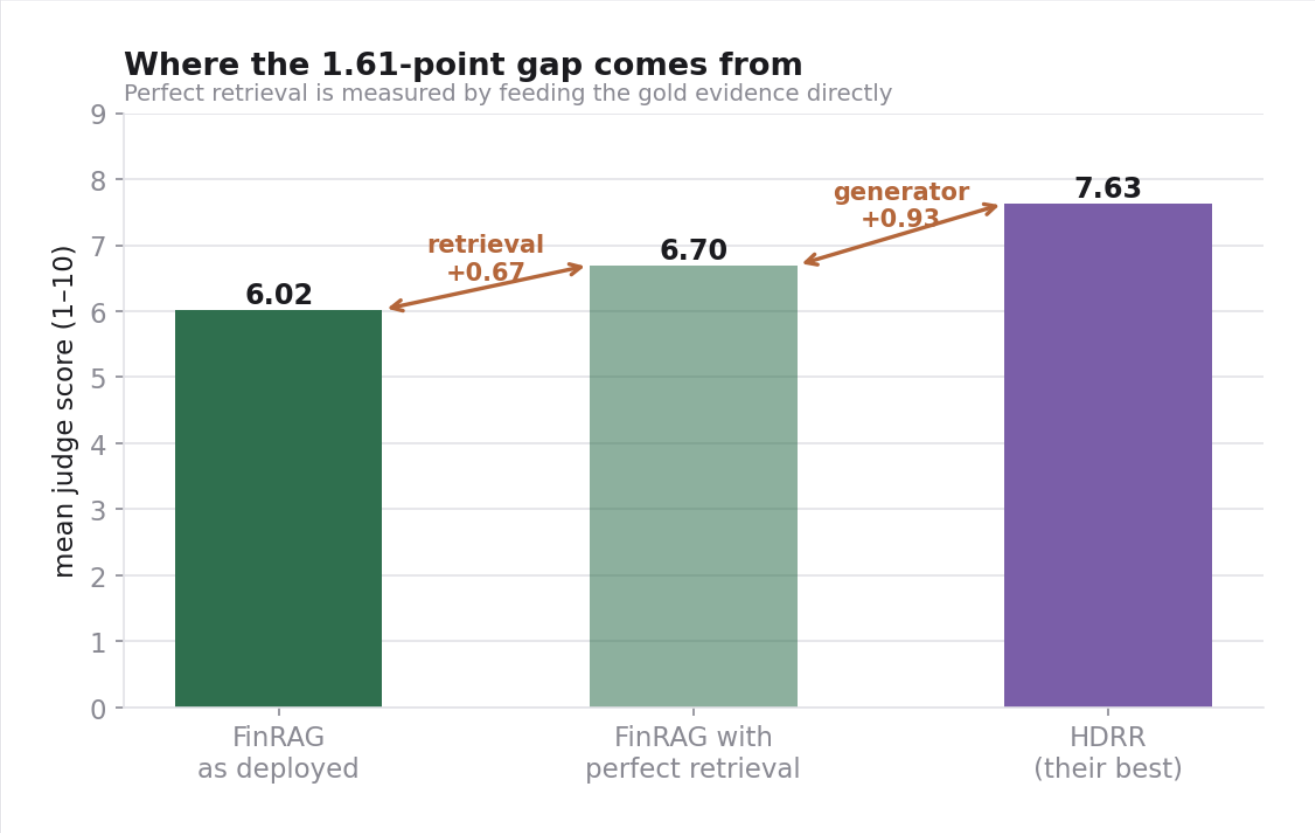
Is the difference real?



Every system is scored on identical queries, so the differences can be resampled directly. FinRAG is statistically tied with Agentic RAG and trails the rest.

FIGURE 3

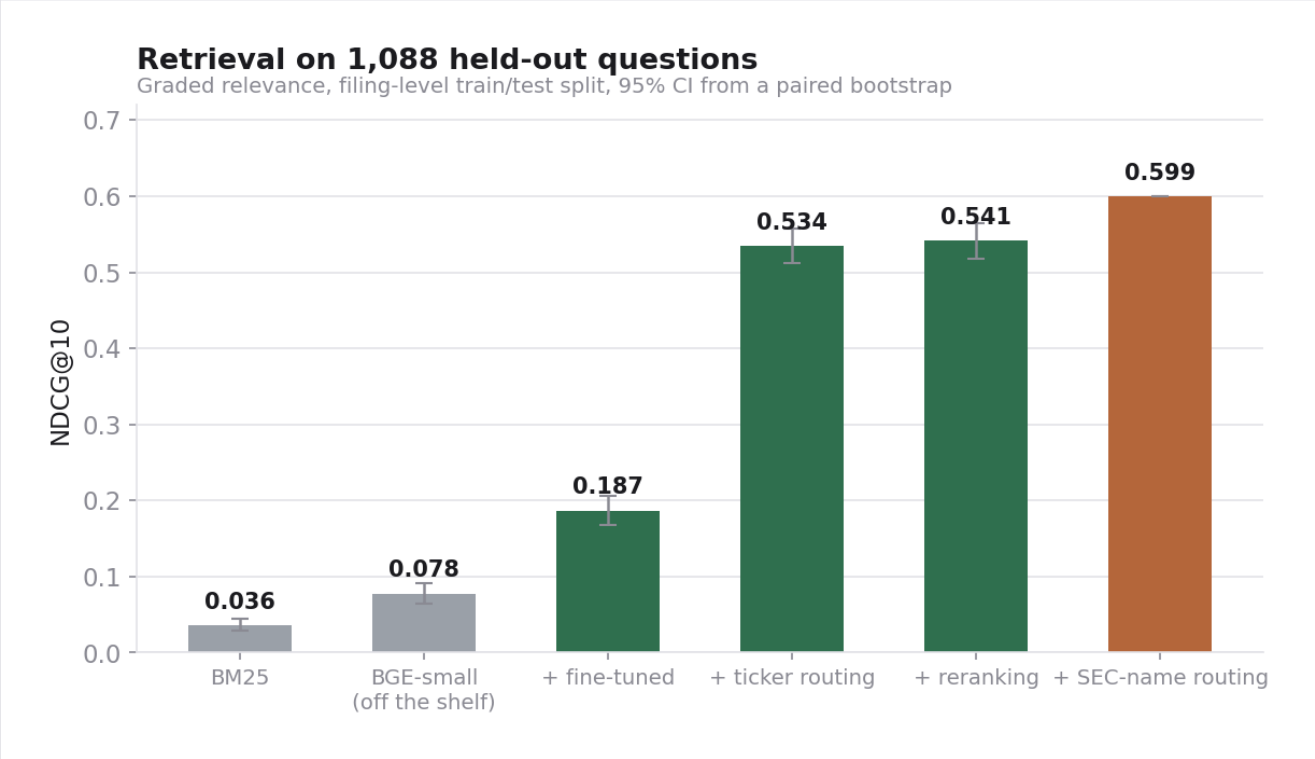
The gap splits into two different problems



0.67 points are lost to retrieval, which is fixable. 0.93 remain even with perfect evidence in front of the model — a 7B model on a laptop against a frontier API model. That is a hardware budget, not a method flaw.

FIGURE 4

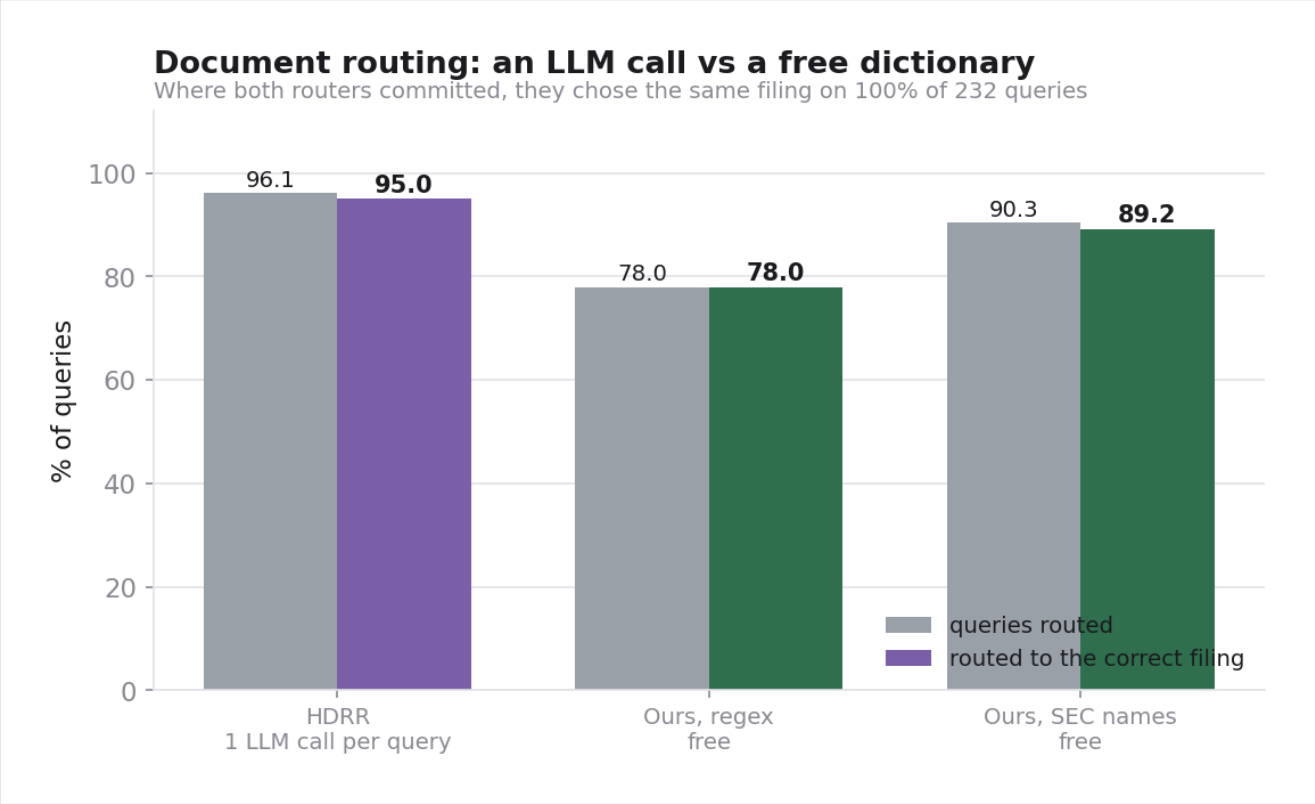
Retrieval — the metric their paper never reports



HDRR publishes only end-to-end judge scores, so there is no published retrieval number on FinDER to compare against. Every step here is measured on filings never seen in training.

FIGURE 5

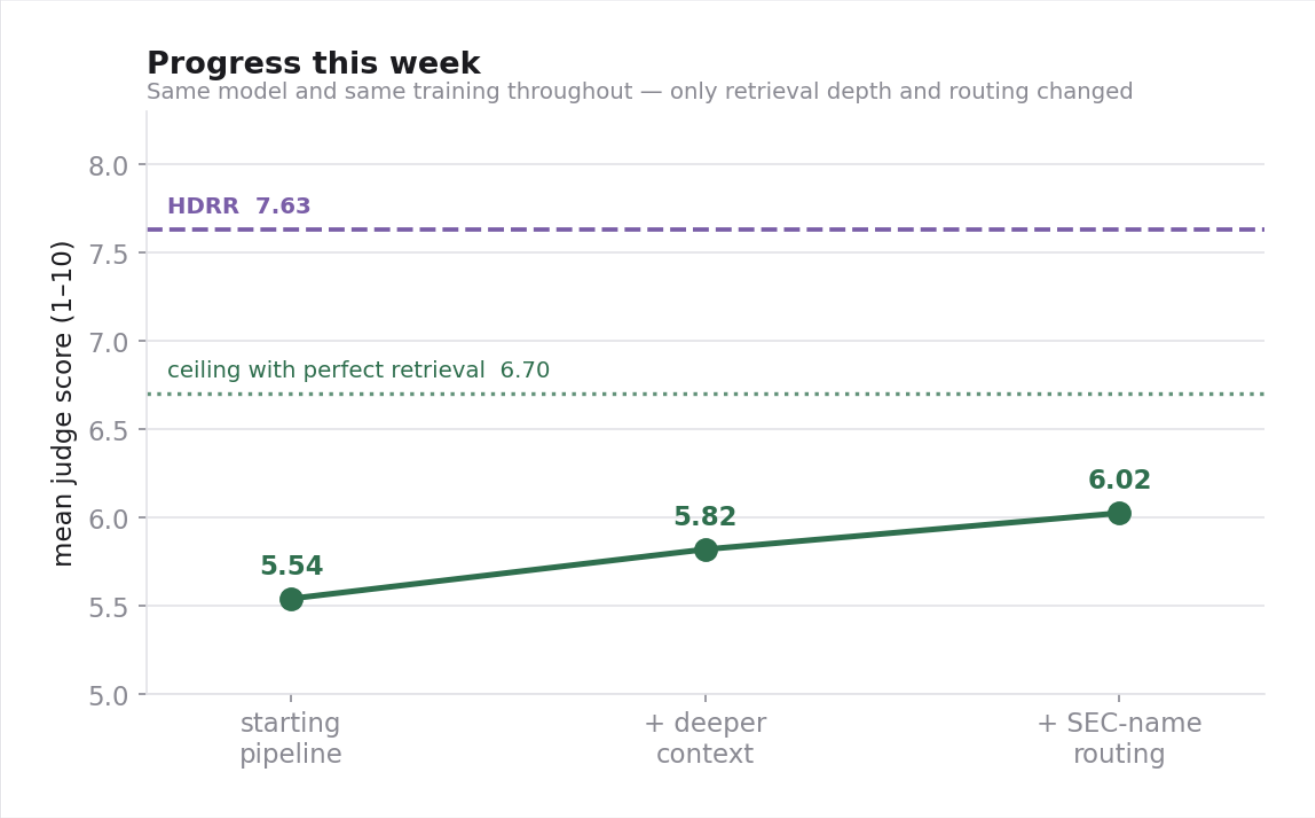
Matching their LLM router at zero cost per query



HDRR calls an LLM on every query to extract the company ticker. Replacing it with SEC's public company-name table plus an acronym stoplist recovers most of that capability and costs nothing to run.

FIGURE 6

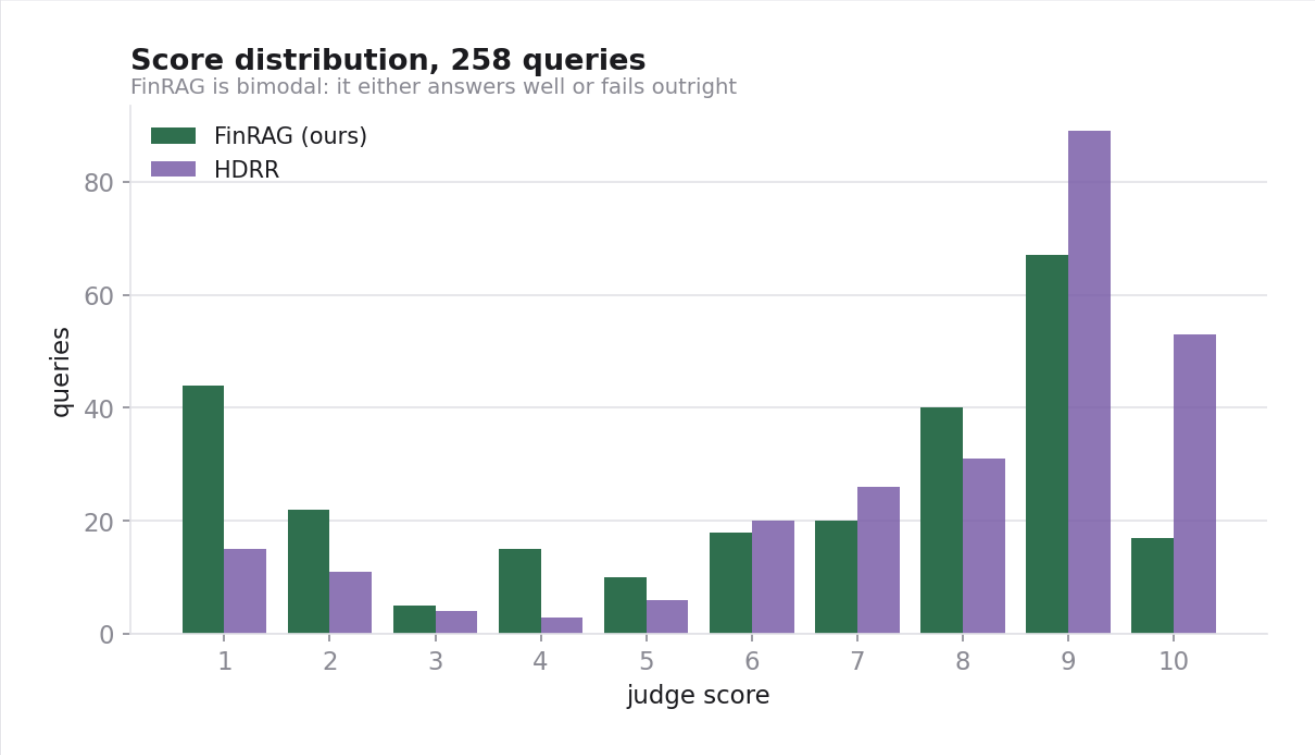
What changed this week



No retraining was involved: 5.54 → 6.02 came from feeding the generator more retrieved context and fixing which company it searches.

FIGURE 7

How the scores are distributed



FinRAG scores 8+ on 48% of queries, close to HDRR's quality when it works. Its losses are concentrated in outright failures rather than mediocre answers — the signature of missing evidence rather than weak reasoning.